

JAYDEN CHEN

jaydencan11@gmail.com, j2692che@uwaterloo.ca | (778) 682-8778 | [linkedin.com/in/jayden-chen](https://www.linkedin.com/in/jayden-chen) | github.com/jchen208208

EDUCATION

University of Waterloo

BASc, Computer Engineering (Honours Co-op)

Sept 2026 – June 2031

Burnaby South Secondary School

• 99%+ average across Grades 11 and 12; Principal's List; 10+ AP courses.

Sept 2020 – June 2026

EXPERIENCES

Lead Student Researcher/First Author — Supervised by Prof. Zhen Bi, Huzhou Normal University

June 2026 – Present

Python, Ollama, Pandas, NumPy, Matplotlib, LaTeX

- Built a multi-agent pipeline automating financial claim verification on the FinDVer benchmark, matching cloud-only accuracy at ~50% of inference cost across 1,700 held-out claims; architected the routing system sending each claim to a local 3B/7B model or cloud API.
- Created two custom verification skills for the system that further cut cloud calls by 20% while it continues to tie cloud-only accuracy.
- Built a BM25 retriever that retrieves 75.2% of gold evidence, 5.4 points above the best published retrieval model on FinDVer.

Research Assistant/Co-author — Under Prof. Zhen Bi, Huzhou Normal University

May 2026 – Present

PyTorch, scikit-learn, NumPy, LlamaGuard/XGuard, Hugging Face Diffusers

- Co-developed TRACE, a text-to-image steering framework triggering sparse-autoencoder steering only at high-risk denoising steps.
- Cut attack success rate by nearly 50% on two test models at ~1% latency overhead, using only 48 MB of added weights.
- Trained 50+ per-layer linear risk probes and selected the most predictable layer by AUPRC score and SAE reconstruction error (FVU).
- Built the projection and mean-pooling, compressing activations 256× (512 KB → 2 KB), enabling a single-pass sweep across all layers.

Research Assistant/Co-author — Under Prof. Zhen Bi, Huzhou Normal University

July 2026 – Present

Python, PyTorch, Hugging Face Transformers, NumPy

- Co-developed C3M, a multimodal memory framework maintaining a 36-entry, 6,144-token active index for long-horizon AI agents.
- Achieved 73%/68% accuracy on the MemLens benchmark at 32K/64K token histories, beating the next best baseline by 9/3%.
- Built the evaluation pipeline that feeds MemLens' 789 questions in session order, testing 5 task types across 3 context lengths.
- Implemented the confidence-thresholded rule deciding if new memories merge, update, or stay separate from existing entries.

IT Technician — Burnaby South Secondary, BC

Sept 2025 – June 2026

- Resolved 100+ hardware and software issues at a ~2,000-student school over 9 months; issues ranged from printer connectivity problems, login and authentication failures, corrupted file recovery, and physical repairs, including broken monitor arms.

Private Tutor

Oct 2024 – Present

- Taught Python and C++ fundamentals alongside math and science to 6+ students in Grades 5–12, breaking down technical concepts for students with no prior programming background.

PROJECTS

Gesture-Controlled Audio Device (SPARC)

May 2026 – Present

Embedded C++, ESP32, Python, PCB Design, CAD Modelling, Soldering

- Developed an embedded device distinguishing hand gestures across two calibrated zones for play/pause, skip, and volume control.
- Designed a custom 2-layer PCB in KiCad and 3D-printed enclosure in Fusion, taking it from a breadboard to a production-ready MVP.
- Confirms every gesture with an animation on an 8-LED strip and streams live track progress via Apple's Media Service over Bluetooth.

Custom Unix Shell

June – Sept 2026

C, CMake, POSIX, termios, Valgrind, Electron, React

- Built a Unix shell in C with custom built-ins, pipelines, redirection, and job control, with leak-free memory management via Valgrind.
- Added a Pokémon-themed animated GUI along with a custom 8-subcommand extension with a local LLM for shell Q&A and lookups.

Tic-Tac-Toe Engine → Neural Network

July – August 2026

C++, Python, PyTorch

- Implemented a depth-weighted minimax algorithm with alpha-beta pruning in C++; unbeatable across all 681 possible game lines.
- Trained an MLP on 4,520 positions generated by the above engine to play optimally without search at ~1,000× lower inference latency.

Minesweeper Neural Network

August 2026 – Present

Python, PyTorch, Pygame

- Building a convolutional neural network that treats board as image, learning number-to-mine patterns and not memorizing positions.
- Built a graphical board interface that shows the network's mine-probability predictions for each cell in real time.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, SQL, HTML/CSS

Frameworks & Tools: Git/GitHub, Next.js (Node.js, React), PostgreSQL, Supabase (Google OAuth), PyTorch, NumPy

Hardware & Design: Soldering, PCB design (KiCad), CAD modelling (Fusion)

Spoken: English (fluent), Mandarin (fluent), French (proficient)